Early Termination of Pandemic Unemployment Insurance and Local Labor Market Recovery: A Comparison of Contiguous Counties Across State Borders

Senior Thesis

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by

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1 Introduction

Unemployment insurance (UI) has long been a core topic in labor economics because it serves two purposes that are conceptually linked but sometimes in tension. On the one hand, it provides temporary income replacement to workers who lose jobs, helping households smooth consumption during periods of economic distress. On the other hand, by altering the value of remaining unemployed relative to accepting a new job, UI may affect job search intensity, reservation wages, and the timing of reemployment. This tradeoff has shaped debates over UI policy across very different economic settings, from ordinary business-cycle downturns to severe crises such as the Great Recession.

Most recently, the COVID-19 pandemic triggered an unprecedented expansion of UI in the United States. Beginning with the 2020 Coronavirus Aid, Relief, and Economic Security (CARES) Act, three major federal programs substantially broadened both eligibility and benefit generosity. Pandemic Unemployment Assistance (PUA) extended UI eligibility to groups traditionally excluded from regular state systems, including self-employed and gig workers; Pandemic Emergency Unemployment Compensation (PEUC) extended benefit duration; and Federal Pandemic Unemployment Compensation (FPUC) added a flat weekly federal supplement, initially $600 and later $300. Together, these programs provided critical income support during a period of severe labor-market disruption. At the same time, they sparked an active debate over whether unusually generous UI benefits discouraged job search and slowed the recovery of employment. For many recipients, total pandemic UI payments exceeded pre-pandemic earnings (Ganong et al., 2020), sharpening the tension between social insurance and labor-supply incentives.

This debate became especially salient in mid-2021, when 21 states chose to terminate the federal pandemic UI programs several months before their scheduled expiration. These early terminations created quasi-experimental policy variation for evaluating the central question: to what extent does withdrawing expanded unemployment benefits accelerate labor-market recovery? Recent studies, including Holzer et al. (2024) and Coombs et al. (2022), find that early termination increased unemployment-to-employment transitions, but much of this evidence relies on state-level comparisons. Such estimates may be confounded by differences in reopening patterns, industrial composition, and regional recovery trajectories that are difficult to fully absorb with aggregate controls alone.

This thesis re-examines the effects of pandemic UI termination using a more geographically localized research design. Following the methodology popularized by Dube et al. (2010), I implement a contiguous border county-pair (CBCP) design that compares neighboring counties on opposite sides of state borders with different UI termination policies. Because adjacent border counties are likely to share similar local economic conditions, this approach helps isolate the effects of benefit termination from broader regional

shocks. I combine this design with two county-level administrative datasets from the U.S. Bureau of Labor Statistics (BLS). The Local Area Unemployment Statistics (LAUS) provide measures of aggregate labor-market adjustment, while the Quarterly Census of Employment and Wages (QCEW) captures establishment counts, employment, and wage bills. Taken together, these data allow me to examine not only worker-side labor-market responses, but also whether any such adjustment translated into measurable business-side changes.

The results do not support the view that early benefit termination produced broad-based gains in employment and business activity. Early benefit termination is associated with a statistically significant decline in the unemployment rate. The LAUS estimates are most consistent with a modest shift from unemployment into employment rather than a withdrawal from the labor force. However, because the stock of unemployment is small relative to the stock of employed workers, this adjustment translates into only limited movement in aggregate employment measures. The QCEW results, meanwhile, do not show robust post-policy improvements in payroll employment, establishment counts, or wages, and some industry-specific estimates exhibit pre-treatment movement that calls for caution in causal interpretation.

The remainder of the thesis proceeds as follows. Section 2 reviews the relevant literature. Section 3 describes the data and sample construction. Section 4 presents the empirical strategy and main results. Section 5 reports robustness checks, including the alternative comparison county-pair sample specification. Section 6 concludes.

2 Literature Review

UI plays a dual role in the economy, providing essential income replacement for displaced workers while acting as an automatic stabilizer during macroeconomic contractions. At the individual level, UI smooths consumption and allows time for job search, but higher benefits can modestly lengthen unemployment spells by raising reservation wages (Landais et al., 2018; Schmieder and von Wachter, 2016). At the macro level, UI sustains aggregate demand during layoffs, potentially preventing the "vicious cycle" of declining consumption and further job losses (Kekre, 2022; Petrosky-Nadeau and Valletta, 2019). Recent evidence indicates that these offsetting effects of UI largely balance one another. In particular, Ganong et al. (2020) demonstrated that the CARES Act resulted in historically high replacement rates, with a median of 145%, yet they found no immediate evidence of aggregate job losses. They argue that the $600 supplement's temporary nature and the scarcity of job openings during lockdowns meant that the consumption-boosting effects of UI likely outweighed the search-disincentive effects.

While the initial expansion of benefits in 2020 allowed researchers to observe the economy's response to

a massive liquidity injection, the mid-2021 early termination across 26 states provided an inverse empirical opportunity. Rather than measuring the effects of benefit extension, this policy shift allowed for a direct evaluation of the labor market's response to a sudden benefit contraction. This quasi-experimental setting is critical for testing whether the search-disincentive effects observed in theory are symmetric when the social safety net is withdrawn.

A prominent strand of this literature utilizes broad, state-level survey data to capture these labor market transitions. Holzer et al. (2024) used Current Population Survey (CPS) microdata to estimate labor-market transitions, employing a difference-in-differences (DiD) framework. They found that ending both FPUC and PUA raised unemployment-to-employment flows by approximately 14 percentage points for prime-age workers. These estimates are broadly consistent across specifications, and align with related evidence from Chatterji and Li (2021), who documented similar employment gains among younger cohorts (age 20-24) following the benefit expiration. Holzer et al. (2024) additionally show that early-exit states experienced faster declines in unemployment rates, though effects on the employment-population ratio were positive but more mixed. At the same time, using the Census Household Pulse Survey, they find that financial well-being deteriorated measurably, with households reporting increased difficulty meeting basic expenses.

Beyond monthly survey cycles, another strand of research utilizes high-frequency private data to track real-time labor market adjustments. Coombs et al. (2022) leveraged bank account data from the fintech app EarnIn to follow a sample of over 16,000 UI recipients. By focusing on this specific population, the authors provide a high-resolution look at the labor supply responses of low-income and credit-constrained individuals, the group most vulnerable to fluctuations in benefit generosity. Their difference-in-differences analysis estimated a 6.8 percentage point increase in employment. However, UI receipt dropped by 36 percentage points, implying that the majority of displaced claimants had not found new employment by that time. On average, individuals in early-exit states lost $2,529 in UI benefits during the summer, but only gained $292 in earnings.

Shifting the focus to the firm-side, Hornstein et al. (2022) compare low-wage and high-wage establishments within the same industry and geography to examine differential recovery patterns. Using Homebase (HB) data on service sector firms, they find that low-wage businesses experienced slower employment recoveries and had to raise wages more aggressively to attract staff. These gaps were significantly larger in periods and places with more generous UI, suggesting that UI may have particularly deterred returns to low-wage jobs. Notably, they point out that in aggregate data, these disincentive effects are largely masked by the concurrent demand stimulus.

Despite the valuable insights provided by these studies, significant questions remain regarding geographic granularity and sectoral inclusiveness.

First, reliance on state-level aggregates (e.g., the CPS) may fail to capture heterogeneous effects within a state. Because labor market conditions can vary drastically between sub-regions, state-level estimates may mask the true intensity of the policy impact at the local level. To address this, I adopt the contiguous county-pair design pioneered by Dube et al. (2010). By shifting the unit of analysis to adjacent counties across state borders, this study ensures that the "control" and "treatment" groups share a localized economic environment, providing a more precise lens for observation.

Second, while high-frequency datasets like EarnIn and Homebase offer real-time insights, their scope is largely limited to credit-constrained individuals and small service-sector firms. The impact on the broader private sector remains largely unobserved. By utilizing the Quarterly Census of Employment and Wages (QCEW) administrative census, this study expands the research scope to include sectors such as Finance and Technical Services. This allows for a more comprehensive understanding of whether the labor supply frictions identified in the service sector propagated through the wider economy or remained localized to specific industry archetypes.

3 Data Sources and Construction of Samples

3.1 Data Sources

To get a high-resolution view of the 2021 UI shock, I leveraged two primary administrative datasets. These sources share a consistent geographic granularity at the county level.

First, I use the Local Area Unemployment Statistics (LAUS) to track resident-based labor market transitions. While many studies rely on the CPS microdata, it is often suppressed at the county level, leading to biased estimates. By contrast, LAUS generates small-area estimates by anchoring CPS data to actual administrative UI claims and payroll records. This model-based approach provides a more suitable county-level coverage. From this, I construct monthly metrics for the employment-to-population ratio, labor force participation rate, and unemployment rate, along with log transformations to calculate labor market supply elasticities.

For industry-specific shifts, I turn to the Quarterly Census of Employment and Wages (QCEW). Because QCEW comes from required tax filings, it is a "near-census" of the American workforce, accounting for almost 98% of all non-farm payrolls. This is better suited to my design over private "high-frequency" apps like Homebase or EarnIn, which suffer from user-selection bias and only represent specific niches. The QCEW allows me to extract quarterly data on average weekly wages (AWW), employment levels, and establishment counts across the full spectrum of NAICS industries.

The integration of these two datasets is particularly useful: LAUS informs us the labor market incentive, while QCEW informs us the establishments' response. This dual-track approach allows for internal cross-validation.

Finally, I merged these panels with the 2020 Decennial Census. By using the official 2020 count as a fixed baseline rather than relying on modeled intercensal estimates, I avoid the measurement error that often comes from population reweighting. These counts serve as the vital component of "weight" in my specification, ensuring that a tiny rural county doesn't skew the results of a larger economic cluster.

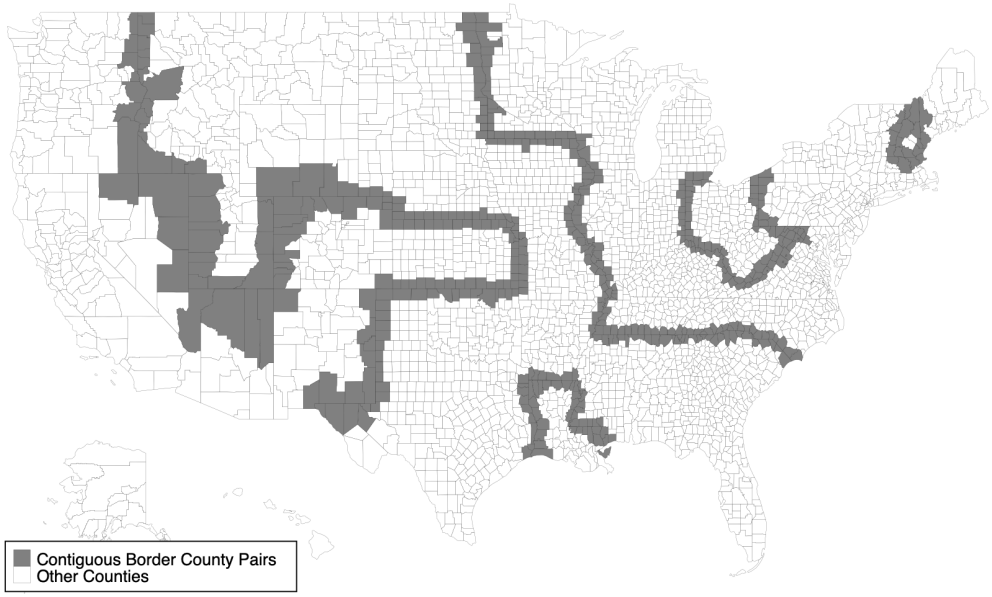
3.2 Sample Construction

To rigorously test the impact of UI termination, I construct three distinct samples that vary in geographic coverage.

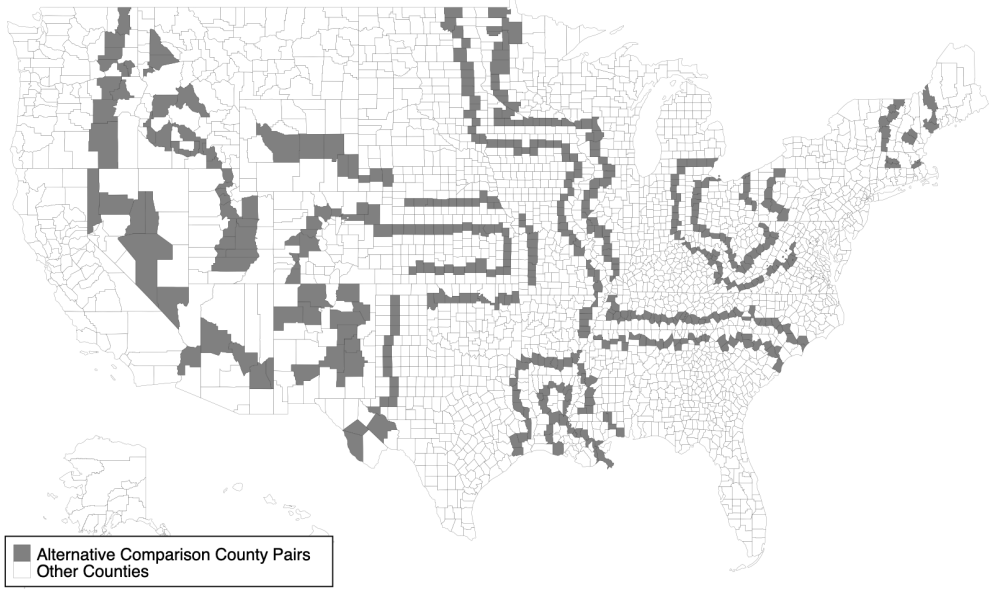
First, to provide a bridge between this study and previous state-level research, I assemble an All-County (AC) sample. This group consists of all the 3,235 counties in the United States. While this sample offers a national perspective, its primary role is for preliminary analysis. To maintain focus on the more precise local analysis, the results derived from the AC sample are included in the Appendix B.

The cornerstone of my empirical strategy is the Contiguous Border County-Pair (CBCP) sample. This design targets counties that straddle state boundaries, ensuring that the treated and control groups are subjected to the same localized economic environment. This main analysis sample consists of 502 distinct county pairs, covering 492 unique counties. Because the regression framework treats each pair as a distinct observation to absorb local shocks, an individual county is replicated in the dataset if it is part of multiple cross-state pairs. This approach allows for a highly granular "within-pair" comparison of labor market outcomes.

Finally, I construct a sample of Alternative Comparison County-Pair, to serve as a robustness check against local spillovers. Ideally, this design would retain the same contiguous county pairs but remove observations within a fixed distance from the state border to eliminate very local spillovers. Because my data are available only at the county level, such distance-based trimming cannot be implemented directly. As a practical approximation, I instead pair counties that lie one county removed from the border. This sample includes 907 distinct pairs across 487 unique counties. By enforcing this physical gap between paired units, the design tests whether the effects observed at the immediate border are driven by hyper-local factors, such as cross-border commuting, or if the policy impact remains consistent when the control group is situated further afield.



(a) Main sample



(b) Robustness check sample

Figure 1: County Pair Construction

Table 1: Descriptive Statistics

	(1) AC sample		(2) CBCP sample		(3) Alternative sample	
	Mean	S.D.	Mean	S.D.	Mean	S.D.
Population, 2020	103,091	331,421	73,577	154,524	70,378	179,401
Population density, 2020	297	1,858	130	339	157	695
Land area (sq. miles)	1,100	3,558	1,193	1,973	1,026	1,475
Overall private employment	35,500	131,689	25,280	68,610	24,483	72,757
Manufacturing employment	3,740	11,393	3,212	6,821	2,983	6,473
Manufacturing weekly wage ($)	923	453	947	422	920	405
Retail employment	4,567	14,352	3,370	7,546	3,249	8,441
Retail weekly wage ($)	540	131	535	107	544	116
Restaurant employment	3,365	11,876	2,446	5,794	2,548	9,957
Restaurant weekly wage ($)	249	164	235	156	238	151
Number of counties	3,235		492		487	
Number of county-pairs	N/A		502		907	
Number of states	20		40		39	

Sources: Quarterly Census of Employment and Wages (QCEW), U.S. Bureau of Labor Statistics; U.S. Bureau of the Census, 2020 Census.

3.3 Contiguous Border Counties as Controls

The core of this research design rests on the premise that contiguous border counties share a unified local labor market, infrastructure, and climate, making them more suitable counterfactuals. By focusing on these adjacent pairs, the model acknowledges that a county is inherently more similar to its neighbor across the border than to a distant county within its own state. This approach avoids the problematic assumption inherent in national or state-level estimates, which implicitly treat every county in the U.S. as an interchangeable and equally valid control for any other. Consequently, these local "twins" allow the regression framework to absorb unobserved local shocks through county-pair-by-period fixed effects, isolating the impact of the policy shift from broader regional trends.

To verify this comparability quantitatively, I perform balance test on pre-treatment period characteristics. The results is presented in Table 2. The first three columns correspond to the CBCP sample, the last three columns report the same statistics for the alternative comparison county-pair sample. Within each sample, the coefficient represents the estimated difference between treated and control counties from a regression of the form $X_{ipt} = \beta \times D_i + \gamma_p + \epsilon_{ipt}$, where D_i indicates treatment status and γ_p denotes county-pair fixed effects. Standard errors are reported in the adjacent column, and the final column reports the mean of the control group.

Across both samples, the estimated differences are generally small relative to their standard errors, indicating no statistically significant differences between treated and control counties in pre-treatment characteristics. In particular, the employment shares and average wage of key industries are highly similar across groups. Furthermore, broader county characteristics such as private-sector employment, population density, and urban share are statistically indistinguishable. Taken together, these results do not reveal systematic pre-treatment differences in observable characteristics between treated and control counties, which supports the credibility of the border-pair identification strategy.

Table 2: Balance Test: Pre-Treatment Treated-Control Comparability

	(1) CBCP sample			(2) Alternative sample		
	β	S.E.	Control Mean	β	S.E.	Control Mean
Manufacturing share	0.014	0.008	0.140	0.017	0.014	0.152
Retail share	0.004	0.007	0.157	0.005	0.004	0.151
Accommodation share	0.003	0.005	0.079	−0.003	0.006	0.081
Manufacturing wage	−47.24	54.55	967.8	−8.067	42.15	878.16
Retail wage	−4.181	13.22	539.0	−11.82	11.78	539.30
Accommodation wage	−11.86	12.97	246.76	−27.88	17.87	238.83
Private sector employment	−4,937	40,283	28,051	−35,003	27,153	24,491
Population density	96.17	206.0	124.0	−196.4	119.0	139.4
Urban share	0.006	0.040	0.356	−0.041	0.038	0.354

Notes: Regressions include county-pair fixed effects. Time t restricted to pre-treatment period. Standard errors are clustered by state. Weight $w :=$ (population / # county appearances). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.4 Choice of Industry

While I maintain access to the full spectrum of NAICS-coded industries, including all sectors in the main analysis would be distracting and redundant. Therefore, I deliberately focus on a selection of representative industries categorized by their expected sensitivity to UI supplements. My primary hypothesis is that labor-intensive service sectors will exhibit the most pronounced response to UI withdrawal, such as Retail (44–45) and Accommodation and Food Services (72). These industries are characterized by high worker turnover and relatively low reservation wages, making them more sensitive to the policy shock. Conversely, I utilize high-skill sectors such as Finance and Insurance (52) and Professional and Technical Services (54) as a structural baseline. Workers in these fields typically possess stronger firm attachment and earn wages that far exceed UI replacement rates, making their employment decisions relatively inelastic to the termination of supplements. Furthermore, their greater remote-work flexibility nature suggests they are less likely to be directly affected.

By focusing on these two divergent groups, the analysis can more clearly isolate the behavioral impact of the policy change from broader macroeconomic fluctuations.

4 Empirical Strategy and Main Results

4.1 Identification Using the Contiguous Border County-Pair Sample

The baseline empirical specification is an event-study regression:

$$Y_{ipt} = \sum_{k \neq 0} \beta_k \cdot D_i \times \mathbb{1}\{t - T = k\} + \alpha_i + \gamma_{pt} + \varepsilon_{ipt} \tag{1}$$

Y_{ipt} denotes the outcome variable for county i belonging to county pair p at time t. Outcomes capture various measures of labor-market activity and business dynamics, which are described in detail in Sections 4.1.1 and 4.1.2. The binary regressor D_i indicates whether county i belongs to a state that terminated pandemic UI benefits early. It equals to 1 if the state the county belongs to ended the benefits earlier than expected, 0 otherwise. The event-time indicators $\mathbb{1}\{t-T=k\}$ measure the number of periods relative to the early termination time T (June 2021), with $k=0$ omitted as the baseline period. The coefficients β_k trace the dynamic evolution of various outcomes in counties located in early-termination states relative to counties in neighboring states that maintained the programs. The pre-treatment coefficients serve as a diagnostic for the identifying parallel-trends assumption underlying the border-county design, while the post-treatment coefficients capture how local labor market respond to the early benefit withdrawal.

The specification includes county fixed effects α_i , which absorb all time-invariant characteristics of each county, such as geographic features, persistent differences in culture, and long-run demographic characteristics. In addition, the regression includes county-pair-by-time fixed effects γ_{pt} , which control for shocks common to both counties within each border pair in a given period. Because counties within each pair are geographically adjacent and likely share similar economic environments, the pair-period fixed effects absorb many local demand shocks, leaving variation driven primarily by differences in state-level UI policy exposure.

All regressions are weighted by county population divided by the number of times a county appears in the stacked county-pair sample (i.e. $w_i = \dfrac{pop_i}{m_i}$ where m_i is the number of pair observations containing county i). Because a given border county may be matched to multiple adjacent counties across the state boundary, the stacking procedure causes some counties to enter the sample repeatedly. This weighting scheme prevents counties with many cross-border neighbors from receiving disproportionate influence purely due to sample construction, while still allowing more populous counties to carry greater weight in proportion to their economic importance.

Standard errors are clustered at the state level to account for potential correlation in unobserved shocks within states. Because the treatment variation in this thesis arises at the state level, counties within the same state may share common shocks or policy-related disturbances that generate correlated regression residuals. State-level clustering therefore provides the appropriate baseline for statistical inference. Since the unit of observation is the county, I also report county-clustered standard errors in Appendix C as a supplementary robustness check.

Because the datasets used in the analysis differ in temporal frequency, the event-study window varies across specifications. Labor-market supply outcomes derived from LAUS are available at a monthly frequency, while business activity measures from the QCEW are available at a quarterly frequency. The specific event windows used for each dataset are described in the following subsections.

4.1.1 Aggregate Labor Market Outcomes: LAUS

I first apply the empirical specification to county-level labor market indicators constructed from the LAUS data. Since it is monthly data, I estimate the event-study specification using an event window that spans five months prior to the policy change and three months afterward (i.e. $k \in [-5, 3]$). The analysis focuses on three key labor market indicators: the unemployment rate (UR), the employment-to-population ratio (Epop), and the labor force participation rate (LFPR). As supplement, the logarithm of raw employment, unemployment, and labor force count are also included.

4.1.2 Local Business Outcomes: QCEW

I next apply the empirical specification to county-level measures of local business activity constructed from the QCEW data. In contrast to the LAUS outcomes, which are available at a monthly frequency, the QCEW variables are reported quarterly. The event-study window spans three quarters prior to the policy change and one quarter afterward (i.e. $k \in [-3, 1]$). The post-treatment horizon is limited to one quarter because, after that point, all counties in the sample are treated and treatment-timing variation is exhausted. The analysis focuses on three indicators of local business activity: the logarithm of establishment counts (lnEC), the logarithm of average weekly wage (lnAWW), and the logarithm of quarterly employment (lnEmp).

4.2 Main Findings

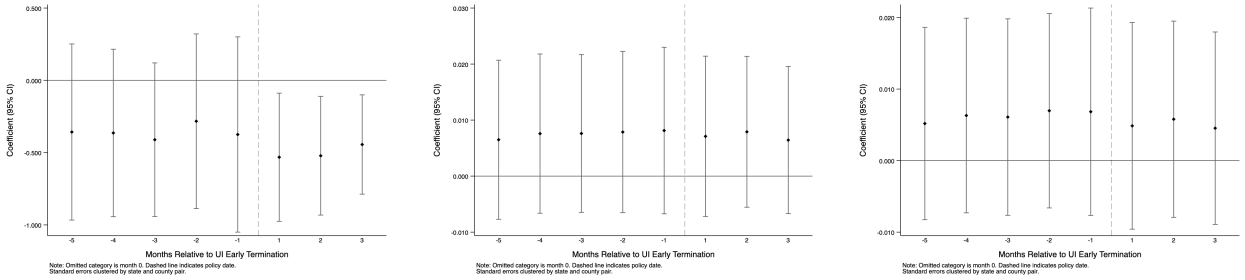
4.2.1 LAUS Outcomes

Figure 2 presents the results of estimating equation (1) with the LAUS outcomes. The precise estimates are presented in Table A1. The estimates indicate a significant decline in the unemployment rate following the early termination of pandemic UI benefits, while the employment-to-population ratio and labor-force participation rate remain unchanged. This pattern appear puzzling at first glance: a reduction in unemployment would typically be expected to coincide with a visible increase in employment or a decrease in labor force participation or both. However, this apparent discrepancy is largely mechanical and reflects the relative sizes of the underlying labor-market stocks.

A back-of-the-envelope calculation helps illustrate this point. The estimated coefficient implies roughly a 0.53 percentage-point decline in the unemployment rate. Applied to the average labor force size in the sample, this corresponds to approximately 190 individuals transitioning out of unemployment in the average county. While this represents a meaningful change relative to the number of unemployed workers, it amounts to only about a 0.56 percent increase in total employment, because the employed population is substantially larger than the unemployed population.

A further caveat is that this estimate may still overstate the size of the unemployment response. As Table A1 shows, the estimated coefficients in the pre-treatment periods are already negative and economically nontrivial in magnitude, even though they are not statistically significant under state-level clustering. This distinction is important because, as emphasized by Rambachan and Roth (2023), failure to reject pre-treatment differences does not by itself validate the parallel-trends assumption. Pre-period estimates can still be informative about the sensitivity of causal conclusions to possible violations of parallel trends. The average of the five pre-treatment coefficients is -0.358, suggesting that treated counties may already have been experiencing somewhat lower unemployment relative to control counties before the policy change. A simple sensitivity check is therefore to compare the post-treatment estimate of -0.532 to this pre-existing gap, which implies a net decline of only about -0.17 percentage points. Under this interpretation, the implied reduction in unemployment is even smaller than the headline estimate suggests. I also report the specification with county-level clustering in Table C1, where the pre-treatment coefficients become statistically significant, further underscoring the need for caution in interpreting the LAUS estimates as clean causal effects.

To conclude, even a sizeable proportional decline in unemployment can translate into only a minimal change in aggregate employment measures such as the employment-to-population ratio. The LAUS results are therefore consistent with a modest shift out of unemployment, but the magnitude is too small relative to the overall workforce to generate a detectable movement in aggregate employment indicators.



(a) Unemployment Rate (b) Employment-to-Population Ratio (c) Labor Force Participation Rate

Figure 2: Event Study Estimates for Worker Labor Market Outcomes (LAUS)

4.2.2 QCEW Outcomes

To evaluate the impact of the policy shock on local business environments, I first establish a baseline using outcomes aggregated across the total private sector (NAICS Industry Code 10). This "All-Industry" benchmark allows for an estimation of the net across-industry effect of the policy shock before exploring a spectrum of industries.

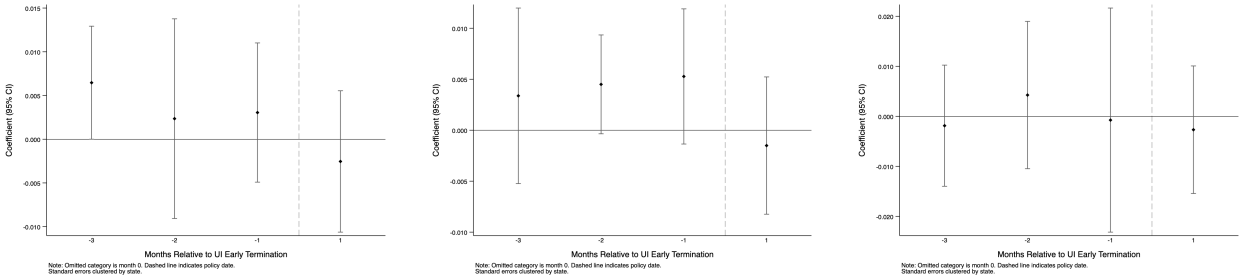
The following sectoral heterogeneity analysis is motivated by a straightforward labor-supply mechanism:

the withdrawal of pandemic UI benefits should not be expected to affect all industries uniformly. To the extent that enhanced UI raised reservation wages and reduced search intensity, the effect should have been strongest in labor-intensive, lower-wage sectors, where jobs are relatively easier to match into, worker turnover is high, and the outside option created by generous benefits is more likely to bind. In such sectors, ending expanded benefits may accelerate job acceptance, producing a sharper short-run adjustment in employment. By contrast, in higher-skill sectors, where matching frictions are higher, vacancies are harder to fill with observationally equivalent workers, and production is more compatible with remote work arrangements, labor demand and employment should be less responsive. This reasoning is consistent with pandemic-era evidence showing that employment losses and subsequent recovery were disproportionately concentrated in low-wage service activities such as retail and leisure and hospitality sectors. (Ganong et al., 2020; Bartik et al., 2020)

To implement the analysis, I group several representative industries into two subsamples. The first group consists Retail (44–45), Administrative and waste services (56), and Accommodation and Food Services (72). These sectors are characterized by relatively low pay, high worker turnover, and limited scope for remote work, making them the industries in which the reservation-wage effects of expanded UI should be most salient. (Rose and Shem-Tov, 2023) By contrast, the second group is composed by Information (51), Finance and Insurance (52), and Professional and Technical Services (54) in which hiring depends more on specialized match quality and where remote or hybrid working mode is substantially greater. (Dingel and Neiman, 2020) This grouping therefore provides a natural empirical contrast between sectors where labor-supply incentives are most likely to bind and sectors where adjustment margins are plausibly weaker.

Figure 3 presents the baseline estimates for local business outcomes aggregated across the total private sector. For all three primary measures (lnEC, lnEmp, and lnAWW) the estimated coefficients in the pre-policy period ($k \in \{-3, -2, -1\}$) are statistically insignificant. This lack of pre-trend significance provides support for the parallel trend assumption, suggesting that treatment and control groups were on comparable trajectories prior to the policy shock. Similarly, the post-policy estimates for $k = 1$ remain insignificant, indicating that the early termination of expanded UI benefits had no discernible net impact on these aggregate business measures. This finding of aggregate stability is consistent with Dube (2021)'s study reporting a limited response to the early withdrawal of pandemic-era UI supplements at the economy-wide level. However, such highly aggregated results may conceal significant but divergent effects across specific industries.

Figure 4 and figure 5 present the event-study estimates for labor-intensive and high-skilled industries, respectively. The estimates do not fully match my expectation. Across both subsamples, there is little evidence of a post-treatment increase in establishment counts or employment. The main visible adjustment

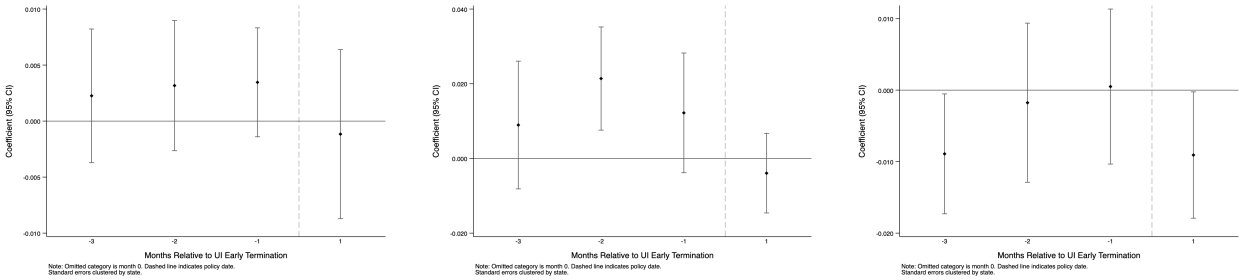


(a) Establishment Count (log) (b) Employment Level (log) (c) Average Weekly Wage (log)

Figure 3: Baseline (All-Industry: 10) Estimates for Local Business Outcomes (QCEW)

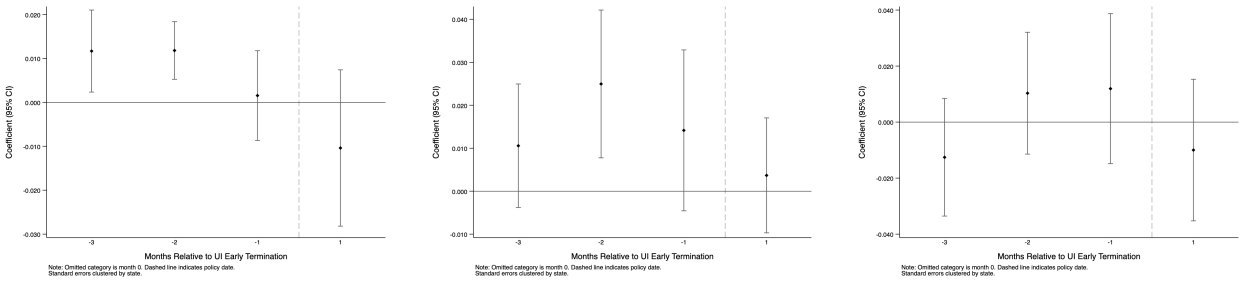
appears in labor-intensive industries, where average weekly wages decline after early termination. One interpretation is that the policy shock affected wage-setting margins more than hiring margins. A possible explanation is that the original hypothesis placed too much weight on labor supply alone: if firms were not expanding labor demand, greater willingness to work would not necessarily raise employment, but could place downward pressure on wages.

These results should nevertheless be interpreted with caution. Both subsamples display positive pre-treatment coefficients in employment, and the high-skilled group also shows a positive pre-trend in establishment counts. Such patterns may partly capture the underlying recovery dynamics that motivated early termination, but they also indicate that the parallel-trends assumption is less secure in these industry-specific specifications. For that reason, the subsample results are best understood as suggestive evidence of sectoral heterogeneity rather than as clean causal estimates.



(a) Establishment Count (log) (b) Employment Level (log) (c) Average Weekly Wage (log)

Figure 4: Labor-Intensive Industries QCEW Outcomes



(a) Establishment Count (log) (b) Employment Level (log) (c) Average Weekly Wage (log)

Figure 5: High-Skilled Industries QCEW Outcomes

5 Robustness Tests

5.1 Using the Alternative Comparison County-Pair Sample

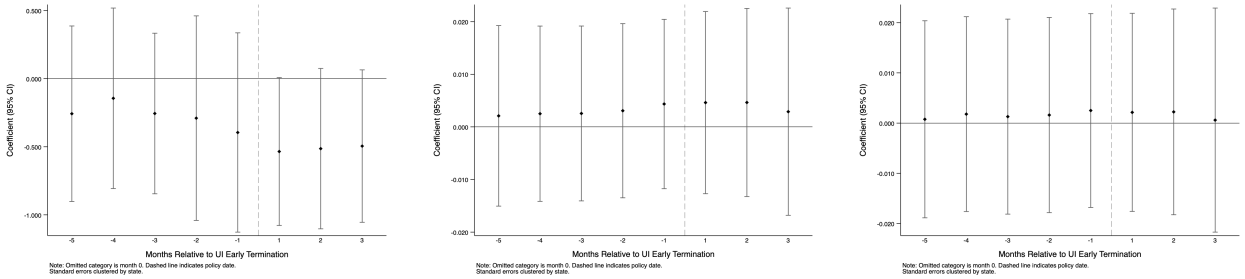
As a robustness check, I re-estimate the baseline design using the alternative comparison county-pair sample. This exercise is intended to assess whether the main results are driven by immediate cross-border spillovers rather than by the policy discontinuity itself. Because the alternative sample shifts the comparison one county inward on each side of the border, it is less exposed to hyper-local border interactions such as cross-border commuting, shopping, or employer responses that may contaminate the contiguous-pair estimates. Ideally, such a spatial robustness test would exclude observations within a fixed distance from the border or use a more granular measure of local economic exposure, but because the available data are observed only at the county level, the alternative pair design provides a feasible county-level approximation.

Relative to the main sample, this design offers a cleaner test against local border contamination, though at the cost of weaker local comparability. Counties located farther from the border may be less tightly linked economically and may differ more in underlying labor-market conditions, so some attenuation in the estimated effects is expected even if the baseline results reflect a genuine policy response. For that reason, the alternative comparison county-pair specification should be viewed as a more conservative spatial test of the main findings.

The empirical framework remains the same as (1), except that the county-pair-by-time fixed effects are now defined over alternative comparison county-pairs rather than contiguous border pairs. Thus, identification continues to come from within-pair differences over time, while the altered sample allows me to examine whether the estimated policy effects persist once the most immediate border comparisons are removed.

5.1.1 LAUS Outcomes

The robustness check using the alternative comparison county-pair sample yields a pattern that is broadly consistent with the main LAUS results. Across the three aggregate labor-market outcomes, the dynamic coefficients preserve the same qualitative shape as in the main specification: unemployment declines after early UI termination, while employment-to-population and labor-force participation exhibit only modest positive movements. Relative to the baseline estimates, however, the confidence intervals are wider and statistical significance is lower, so the post-treatment effects are more muted. This attenuation is consistent with the logic of the robustness design. By moving one county inward from the border, the alternative comparison county-pair sample is less exposed to immediate cross-border spillovers, but it also sacrifices some of the sharp local comparability that underpins the main border-county design. As a result, the robustness exercise does not overturn the main conclusion; rather, it supports the previous findings are not just an artifact of immediate border contamination.



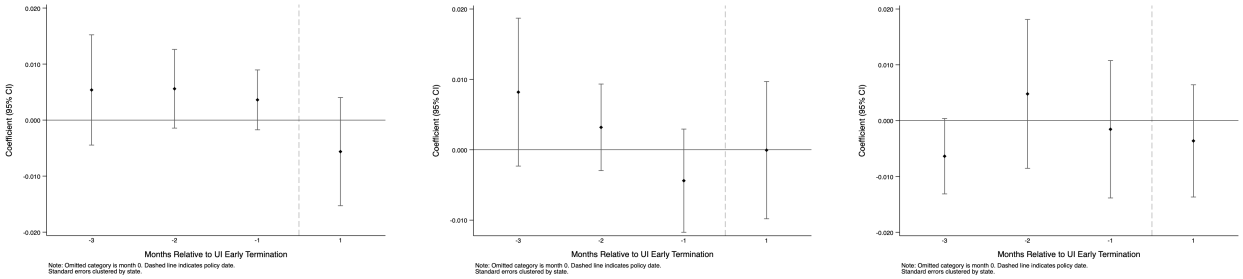
(a) Unemployment Rate (b) Employment-to-Population Ratio (c) Labor Force Participation Rate

Figure 6: Robustness Check for Worker Labor Market Outcomes (LAUS)

5.1.2 QCEW Outcomes

Figure 7 presents the robustness estimates for all-industry QCEW outcomes. The results closely mirror the baseline specification in Figure 3. Across establishment counts, employment, and average weekly wages, the estimated coefficients remain modest and statistically insignificant throughout the event window, with no evidence of a discrete post-treatment break. In this sense, the robustness exercise reinforces the main conclusion that the early termination of pandemic UI did not generate a discernible aggregate response in local business activity at the all-industry level.

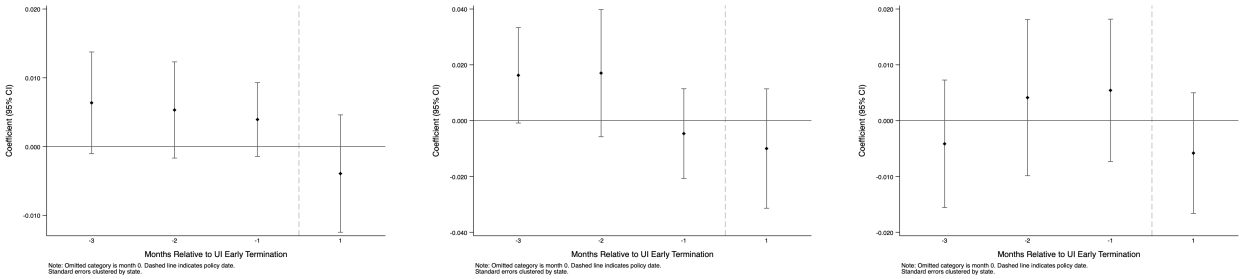
Figures 8 and 9 extend the robustness exercise to the labor-intensive and high-skilled subsamples. The results broadly mirror the main analysis, although the estimates are further muted and less precise. In labor-intensive industries, the post-treatment coefficients again fail to show the expected increase in establishment counts or employment, and the negative wage response present in the CBCP specification becomes



(a) Establishment Count (log) (b) Employment Level (log) (c) Average Weekly Wage (log)

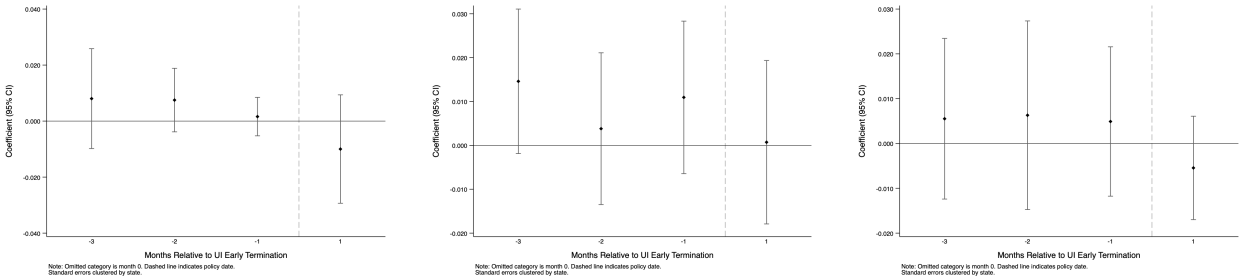
Figure 7: Baseline (All-Industry: 10) Robutness Check for Local Business Outcomes (QCEW)

insignificant. The alternative comparison county-pair estimates therefore leave the interpretation unchanged: early UI termination did not produce a meaningful expansion in labor-intensive business activity, even in the sectors where labor-supply incentives were expected to respond most strongly. In high-skilled industries, the coefficients remain statistically insignificant across all outcomes, which is consistent with the view that these sectors were less responsive to the policy change.



(a) Establishment Count (log) (b) Employment Level (log) (c) Average Weekly Wage (log)

Figure 8: Labor-Intensive Industries QCEW Outcomes Robustness Check



(a) Establishment Count (log) (b) Employment Level (log) (c) Average Weekly Wage (log)

Figure 9: High-Skilled Industries QCEW Outcomes Robustness Check

5.2 Most Affected Industry

Since the predefined labor-intensive and high-skilled groupings do not reconcile cleanly with the aggregate LAUS patterns, I next estimate industry-specific employment responses across all 2-digit NAICS sectors. This exercise is intended to identify whether the aggregate employment adjustment was concentrated in particular sectors, or instead reflected a broader reallocation of workers across industries.

$$\log\left(\text{Emp}_{ipt}^{j}\right) = \beta_j \left(D_i \times \text{Post}_t\right) + \phi_i + \lambda_{pt} + \varepsilon_{ipt}^{j} \tag{2}$$

I estimate equation (2) separately for each 2-digit NAICS industry. The coefficient of interest, β_j, captures the relative post-termination employment response in sector j for treated counties compared with their cross-border controls.

The results in Figure 10 do not support the hypothesis that early termination primarily expanded employment in labor-intensive sectors. Accommodation and food services, administrative and waste services, and construction show negative point estimates, while several white-collar sectors show small positive but statistically insignificant coefficients. Although most estimates are indistinguishable from zero, their pattern is still informative about where the aggregate employment adjustment was unlikely to have occurred. Still, the pattern does not suggest that the aggregate employment improvement in the LAUS data was driven by a surge in low-wage service employment. A more plausible mechanism is sectoral reallocation: workers who had temporarily relied on labor-intensive jobs during the disrupted recovery may have moved back into managerial, technical, or other higher-quality matches once labor-market conditions improved. In that sense, early UI termination may have affected the timing and composition of re-employment more than the scale of hiring within the sectors most exposed to low-skill labor supply.

6 Conclusion and Discussion

This thesis examines whether the early termination of federal pandemic unemployment insurance programs in mid-2021 accelerated local labor-market recovery. Using a contiguous border county-pair design, it compares neighboring counties across state borders that were exposed to different UI termination policies. By restricting comparisons to adjacent counties, the design aims to isolate the effect of policy withdrawal from broader regional differences in recovery, reopening patterns, and industrial composition.

The results do not support a strong version of the labor-supply incentive account, under which ending expanded UI would quickly raise employment, boost payroll activity, and improve local business outcomes. For the aggregate labor market outcomes, early termination is associated with a decline in the unemployment

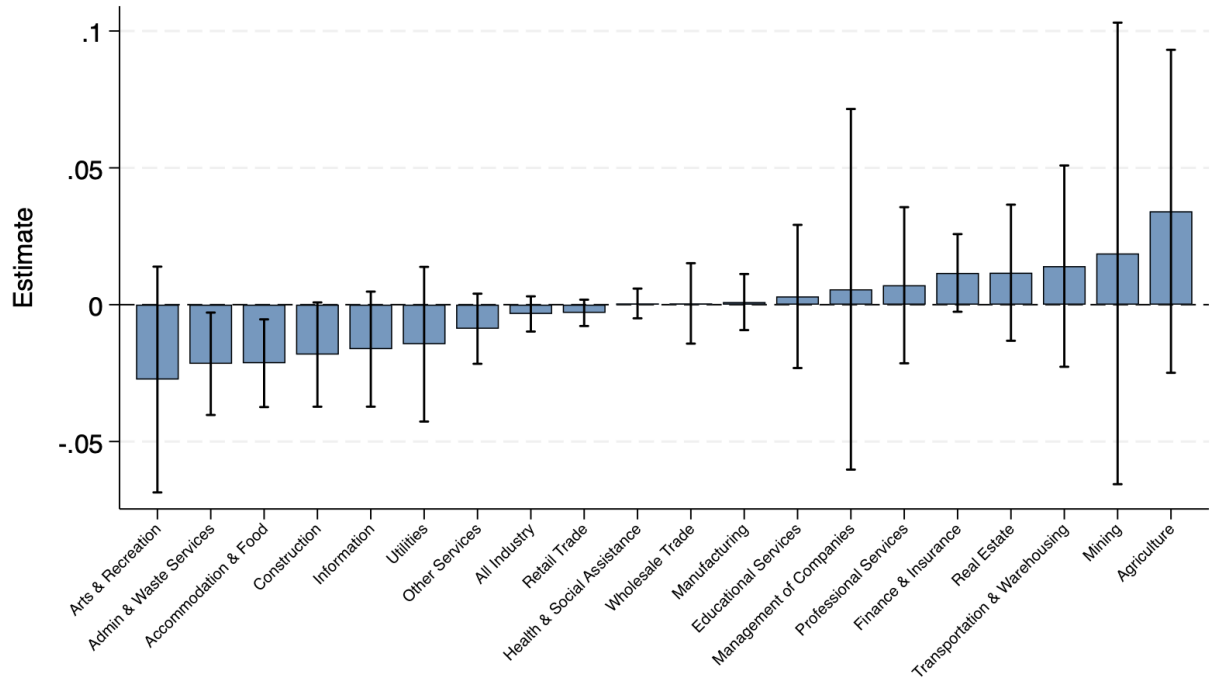


Figure 10: Estimated Employment Effects of Early UI Termination by Industry

rate, but not with a comparably visible increase in the employment-to-population ratio or labor-force participation rate. This pattern is consistent with some workers moving out of unemployment and into employment, while the aggregate effect on broader employment measures remains mechanically small because the stock of employed workers is much larger than the stock of unemployed workers. In this respect, the worker-side results are broadly consistent with prior studies that find stronger unemployment-to-employment transitions following the withdrawal of pandemic UI benefits, including Holzer et al. (2024) and Coombs et al. (2022).

For the business activity outcomes, however, the evidence is more limited. In the QCEW data, there is no meaningful post-treatment expansion in aggregate establishment counts, employment, or average weekly wages. The robustness exercise using alternative comparison county-pairs leaves that conclusion broadly unchanged. This pattern is more closely aligned with Dube (2021), who also finds limited aggregate employment effects from the expiration of federal pandemic UI supplements. Taken together, the results suggest that worker-side adjustment following benefit withdrawal did not translate into a broad-based expansion in local business activity over the horizon observed here.

The sectoral evidence provides a further test of the labor-supply mechanism. If the main effect of expanded UI operated through reduced search intensity or higher reservation wages, labor-intensive industries would be expected to show the clearest post-termination rebound. The estimates do not support that prediction.

In the main QCEW specification, labor-intensive sectors show no post-treatment increase in establishment counts or employment, and wages decline instead. The 2-digit industry exercise points in the same direction: accommodation and food services, administrative and waste services, and construction do not emerge as the sectors driving the aggregate employment adjustment suggested by the LAUS results.

This sectoral pattern is difficult to reconcile with a pure labor-supply interpretation. One possible explanation is that labor demand remained weak in the sectors most exposed to the policy change. Under this interpretation, benefit withdrawal may have increased willingness to accept available jobs without generating substantial new hiring because firms were not expanding labor demand at the same pace. In that case, the immediate adjustment margin would be wage pressure rather than employment growth. The negative wage response in the labor-intensive group is consistent with this interpretation, although the present design does not directly test the demand channel. More broadly, this interpretation is informed by work such as Karahan et al. (2019), which emphasizes that employment dynamics depend not only on worker-side incentives but also the search behavior and the job finding rate per unit of search.

A second interpretation is cross-sector reallocation. During the disrupted recovery, some displaced workers may have entered labor-intensive sectors as temporary fallback options rather than as a durable source of re-employment. If early termination accelerated movement out of those sectors and into other parts of the labor market, then labor-intensive industries would not necessarily absorb the employment adjustment associated with benefit withdrawal and could instead lose part of that temporary workforce. Under this view, early UI termination affected the composition and timing of re-employment more than the aggregate scale of hiring. This is broadly consistent with evidence presented in Section 5.2 and Hornstein et al. (2022), who show that pandemic labor-market adjustment differed across local industry markets and cannot be fully understood through aggregate employment changes alone.

These conclusions should nevertheless be interpreted cautiously. First, the QCEW is quarterly, while the policy variation occurs over a short 3-month window in mid-2021, leaving limited scope to detect delayed or gradual business responses. Second, the LAUS county measures are heavily model-based and imputed, so the worker-side results are necessarily less direct than fully observed administrative counts. Third, some industry-level QCEW specifications exhibit pre-treatment movement, which weakens a strict causal interpretation of the sectoral estimates. For these reasons, the strongest conclusion of this thesis is not that early termination had no effect, but that its observable effects were modest, uneven, and more consistent with limited reallocation than with a sharp, low-wage hiring boom.

These limitations also suggest several directions for future research. A useful next step would be to use data with finer temporal resolution to better capture delayed employer adjustment following benefit withdrawal. Additional evidence on vacancies, hires, hours, job-to-job transitions, or occupation-specific

employment would also help distinguish between competing interpretations of the sectoral results. More broadly, future work could examine whether the effects of early UI termination varied systematically across industries and local labor markets with different recovery conditions, reopening trajectories, and demand environments.

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A Result Tables

Table A1: LAUS Event Study on CBCP Sample Results

	(1) Unemp. Rate	(2) E-Pop	(3) Labor Force Participation Rate
$k = -5$	−0.357	0.006	0.005
	(0.301)	(0.007)	(0.007)
$k = -4$	−0.364	0.008	0.006
	(0.286)	(0.007)	(0.007)
$k = -3$	−0.411	0.008	0.006
	(0.263)	(0.007)	(0.007)
$k = -2$	−0.283	0.008	0.007
	(0.299)	(0.007)	(0.007)
$k = -1$	−0.375	0.008	0.007
	(0.334)	(0.007)	(0.007)
$k = 1$	−0.532**	0.007	0.005
	(0.219)	(0.007)	(0.007)
$k = 2$	−0.521**	0.008	0.006
	(0.203)	(0.007)	(0.007)
$k = 3$	−0.444**	0.006	0.005
	(0.170)	(0.006)	(0.007)
Constant	5.554***	0.433***	0.458***
	(0.002)	(0.000)	(0.000)
F-statistic (pre-trend)	1.696	1.741	1.263
Observations	430,636	430,636	430,636
R^2	0.949	0.936	0.933

Notes: Standard errors in parentheses. All regressions include county and pair-period fixed effects.Standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A2: LAUS Event Study on Alternative Comparison County-Pair Sample Results

	(1) Unemp. Rate	(2) E-Pop	(3) Labor Force Participation Rate
$k = -5$	−0.257	0.002	0.001
	(0.318)	(0.008)	(0.010)
$k = -4$	−0.144	0.003	0.002
	(0.327)	(0.008)	(0.010)
$k = -3$	−0.256	0.003	0.001
	(0.291)	(0.008)	(0.010)
$k = -2$	−0.290	0.003	0.002
	(0.371)	(0.008)	(0.010)
$k = -1$	−0.395	0.004	0.003
	(0.361)	(0.008)	(0.010)
$k = 1$	−0.535*	0.005	0.002
	(0.268)	(0.009)	(0.010)
$k = 2$	−0.514*	0.005	0.002
	(0.291)	(0.009)	(0.010)
$k = 3$	−0.495*	0.003	0.001
	(0.277)	(0.010)	(0.011)
Constant	5.562***	0.431***	0.456***
	(0.002)	(0.000)	(0.000)
F-statistic (pre-trend)	3.240	0.486	1.302
Observations	777,926	777,926	777,926
R^2	0.952	0.952	0.948

Notes: Standard errors in parentheses. All regressions include county and pair-period fixed effects. Standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A3: QCEW Event Study on CBCP Sample Labor-Intensive Group

	(1) log(Estab. Count)	(2) log(Emp)	(3) log(avg weekly wage)
$k = -3$	0.002	0.010	−0.009**
	(0.003)	(0.008)	(0.004)
$k = -2$	0.003	0.022***	−0.002
	(0.003)	(0.007)	(0.006)
$k = -1$	0.004	0.012	0.001
	(0.002)	(0.008)	(0.005)
$k = 1$	−0.001	−0.004	−0.009**
	(0.004)	(0.005)	(0.004)
Constant	6.033***	8.830***	6.364***
	(0.000)	(0.001)	(0.001)
F-statistic (pre-trend)	0.991	1.793	3.845
Observations	24,003	19,434	19,434
R^2	0.935	0.264	0.921

Notes: Standard errors in parentheses. All regressions include county and pair-period fixed effects. Standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A4: QCEW Event Study on CBCP Sample High-Skilled Group

	(1) log(Estab. Count)	(2) log(Emp)	(3) log(avg weekly wage)
$k = -3$	0.012**	0.011	−0.012
	(0.005)	(0.007)	(0.010)
$k = -2$	0.012***	0.025***	0.011
	(0.003)	(0.009)	(0.011)
$k = -1$	0.001	0.014	0.012
	(0.005)	(0.009)	(0.013)
$k = 1$	−0.010	0.004	−0.010
	(0.009)	(0.007)	(0.013)
Constant	5.417***	7.630***	7.271***
	(0.001)	(0.001)	(0.002)
F-statistic (pre-trend)	5.124	2.711	3.913
Observations	23,876	19,067	19,067
R^2	0.791	0.805	0.880

Notes: Standard errors in parentheses. All regressions include county and pair-period fixed effects. Standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Estimated Employment Effects of Early UI Termination by Industry

Industry	(1) β	(2) S.E.	(3) p-value
Arts & Recreation	−0.027	(0.021)	0.201
Admin & Waste Services	−0.022**	(0.010)	0.029
Accommodation & Food	−0.021**	(0.008)	0.013
Construction	−0.018*	(0.010)	0.068
Information	−0.016	(0.011)	0.138
Utilities	−0.014	(0.014)	0.323
Other Services	−0.009	(0.007)	0.186
All Industry (Overall)	−0.003	(0.003)	0.308
Retail Trade	−0.003	(0.002)	0.230
Health & Social Assistance	0.000	(0.003)	0.871
Wholesale Trade	0.000	(0.007)	0.951
Manufacturing	0.001	(0.005)	0.856
Educational Services	0.003	(0.013)	0.823
Management of Companies	0.006	(0.034)	0.868
Professional Services	0.007	(0.015)	0.628
Finance & Insurance	0.012	(0.007)	0.117
Real Estate	0.012	(0.013)	0.363
Transportation & Warehousing	0.014	(0.019)	0.457
Mining	0.019	(0.043)	0.666
Agriculture	0.034	(0.030)	0.265

Regressions include county and pair-period fixed effects. Standard Error clustered by state. * p<0.10, ** p<0.05, *** p<0.01.

B Specifications Using the All Counties Sample

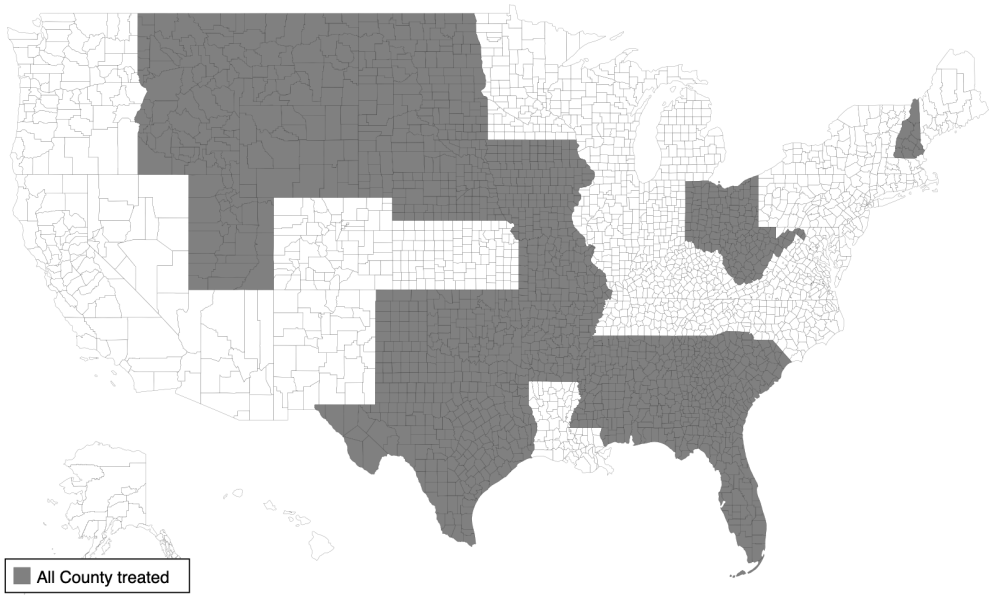


Figure B1: All-Counties Sample

B.1 Balance Test

Table B1: Balance Test for All-County Sample

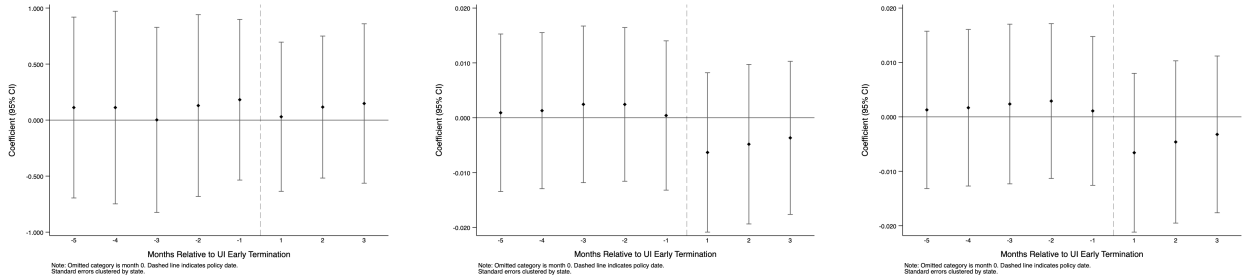
	(1) β	(2) S.E.	(3) Control Mean
Manufacturing share	0.002	(0.005)	0.153
Retail share	0.014***	(0.004)	0.154
Accommodation share	0.014***	(0.003)	0.085
Manufacturing wage	−163.226***	(53.504)	975.6
Retail wage	−64.662***	(20.47)	551.8
Accommodation wage	−69.556***	(14.274)	277
Private sector employment	−204,068	(168,665)	45,201
Population density	−2166.72***	(819.3)	463.9
Urban share	−0.06***	(0.022)	0.426

Notes: Standard errors are robust. Weighted by county population. * p<0.10, ** p<0.05, *** p<0.01

Table B1 confirms that the all-county sample is poorly balanced and therefore provides a weak comparison between treated and control states. Relative to the CBCP and alternative comparison county-pair samples, the treated and control groups here differ substantially in pre-treatment industrial composition, wage levels, population density, and urbanization, with many of these gaps statistically significant. This suggests that, in the full national sample, treated states were not comparable to control states even before the policy change. As a result, any state-level estimate from this sample would be more vulnerable to contamination

by underlying cross-state differences rather than isolating the effect of early UI termination itself. In this sense, the balance test provides direct support for the contiguous-county design: by restricting comparisons to geographically adjacent counties, the preferred samples deliver a more credible counterfactual and a cleaner identification strategy.

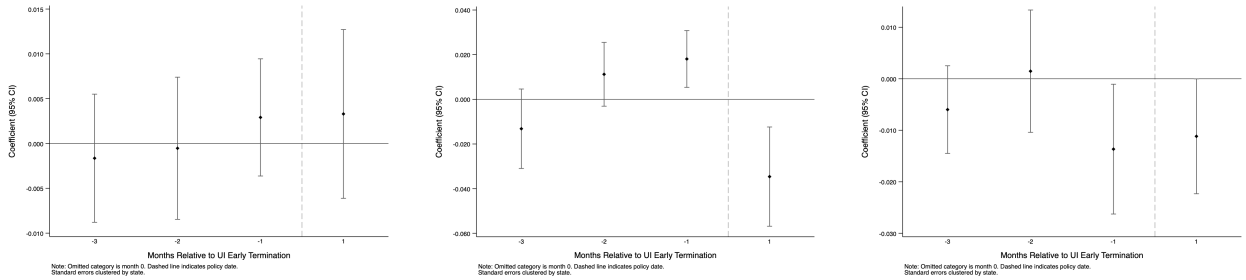
B.2 LAUS Outcomes



(a) Unemployment Rate (b) Employment-to-Population Ratio (c) Labor Force Participation Rate

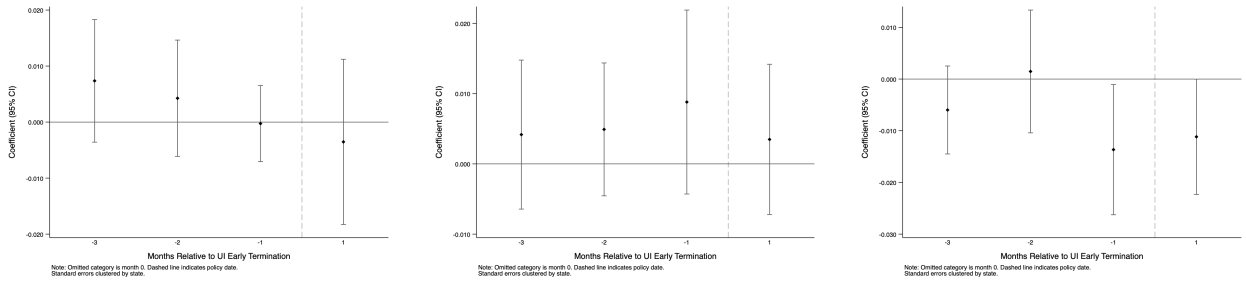
Figure B2: AC Sample Estimates for Worker Labor Market Outcomes (LAUS)

B.3 QCEW Outcomes



(a) Establishment Count (log) (b) Employment Level (log) (c) Average Weekly Wage (log)

Figure B3: Estimate Labor-Intensive Industries QCEW Outcomes on AC Sample



(a) Establishment Count (log) (b) Employment Level (log) (c) Average Weekly Wage (log)

Figure B4: Estimate High-Skilled Industries QCEW Outcomes on AC Sample

C County-level Clustering

Table C1: LAUS Event Study on CBCP

	(1) Unemp. Rate	(2) E-Pop	(3) Labor Force Participation Rate
$k = -5$	-0.357**	0.006	0.005
	(0.173)	(0.005)	(0.005)
$k = -4$	-0.364**	0.008	0.006
	(0.158)	(0.005)	(0.005)
$k = -3$	-0.411***	0.008	0.006
	(0.149)	(0.005)	(0.005)
$k = -2$	-0.283*	0.008	0.007
	(0.165)	(0.005)	(0.005)
$k = -1$	-0.375**	0.008	0.007
	(0.175)	(0.006)	(0.006)
$k = 1$	-0.532***	0.007	0.005
	(0.139)	(0.006)	(0.006)
$k = 2$	-0.521***	0.008	0.006
	(0.125)	(0.005)	(0.006)
$k = 3$	-0.444***	0.006	0.005
	(0.116)	(0.005)	(0.006)
Constant	5.554***	0.433***	0.458***
	(0.001)	(0.000)	(0.000)
F-statistic (pre-trend)	4.196	2.433	2.152
Observations	430,636	430,636	430,636
R^2	0.949	0.936	0.933

Notes: Standard errors in parentheses. All regressions include county and pair-period fixed effects. Standard errors clustered by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C2: QCEW Event Study on CBCP Sample (Baseline)

	(1) log(Estab. Count)	(2) log(Emp)	(3) log(avg weekly wage)
$k = -3$	0.005***	0.002	-0.007
	(0.002)	(0.003)	(0.007)
$k = -2$	0.004**	0.011***	0.011
	(0.002)	(0.004)	(0.009)
$k = -1$	0.004**	-0.002	0.014**
	(0.002)	(0.006)	(0.006)
$k = 1$	-0.001	-0.006**	-0.017**
	(0.002)	(0.003)	(0.008)
Constant	5.140***	6.908***	7.878***
	(0.000)	(0.000)	(0.001)
F-statistic (pre-trend)	3.565	3.448	2.744
Observations	154,058	114,795	114,795
R^2	0.418	0.195	0.453

Notes: Standard errors in parentheses. All regressions include county and pair-period fixed effects. Standard errors clustered by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C3: QCEW Event Study on CBCP Sample Labor-Intensive Group

	(1) log(Estab. Count)	(2) log(Emp)	(3) log(avg weekly wage)
$k = -3$	0.002	-0.009**	0.010
	(0.003)	(0.004)	(0.008)
$k = -2$	0.003	-0.002	0.022***
	(0.003)	(0.006)	(0.007)
$k = -1$	0.004	0.001	0.012
	(0.002)	(0.005)	(0.008)
$k = 1$	-0.001	-0.009**	-0.004
	(0.004)	(0.004)	(0.005)
Constant	6.033***	6.364***	8.830***
	(0.000)	(0.001)	(0.001)
F-statistic (pre-trend)	0.991	1.793	3.845
Observations	24,003	19,434	19,434
R^2	0.935	0.264	0.921

Notes: Standard errors in parentheses. All regressions include county and pair-period fixed effects. Standard errors clustered by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C4: QCEW Event Study on CBCP Sample High-Skilled Group

	(1) log(Estab. Count)	(2) log(Emp)	(3) log(avg weekly wage)
$k = -3$	0.012**	-0.012	0.011
	(0.005)	(0.010)	(0.007)
$k = -2$	0.012***	0.011	0.025***
	(0.003)	(0.011)	(0.009)
$k = -1$	0.001	0.012	0.014
	(0.005)	(0.013)	(0.009)
$k = 1$	-0.010	-0.010	0.004
	(0.009)	(0.013)	(0.007)
Constant	5.417***	7.271***	7.630***
	(0.001)	(0.002)	(0.001)
F-statistic (pre-trend)	5.124	2.711	3.913
Observations	23,876	19,067	19,067
R^2	0.791	0.805	0.880

Notes: Standard errors in parentheses. All regressions include county and pair-period fixed effects. Standard errors clustered by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$